Assignment 1: Localization and Simulation Foundations

Timing: Released Week 2. Submitted via Canvas at the end of Week 4.
Weight: 10% of module mark
Mode: Individual Canvas Submission

1. Overview and Software Support

In this assignment, you will implement and evaluate robot localization in simulation and use local Git version control to record your code development.

You may use a simulation environment of your choice, such as ROS 2/Gazebo, Webots, MATLAB/Simulink, Python/Pygame or PyBullet.

Software support: ROS 2 and Gazebo are available and supported on the G04 laboratory workstations. If you use your own computer, you are responsible for installing and configuring the software and resolving any device or operating system issues. Teaching staff can help with localization concepts and software use in the laboratory but cannot provide installation support for personal computers.

2. The Task: Localization

Implement a method of localization to estimate a mobile robot's position and orientation as it moves through an indoor or outdoor simulated environment.

Steps:

1. Set up a simulation environment containing a mobile robot. Model uncertainty in the robot's odometry and in any sensor measurements used for localization.

2. Implement a localization method to estimate the robot's poses along a short trajectory. You may use odometry or dead reckoning with error tracking, Markov localization or a particle filter. Choose a method suitable for your simulation environment. More complex methods do not automatically receive higher marks.

3. Compare your estimated pose with the simulator's ground-truth pose and analyses the differences. Use ground truth only to evaluate your results; do not use it directly to generate the localization estimate.

4. Use local Git version control to record your code development.

3. Required Deliverables

Submit one ZIP file via Canvas containing the following three components:

- **1-Page Report (PDF):** Include the following sections:
    - **Problem Statement:** A brief definition of the specific localization scenario being tackled (e.g., "Tracking a differential-drive robot in a 2D indoor environment using noisy odometry").
    - **Methods:** A summary of the chosen simulation platform and the mathematical/logical method used to develop the localization estimate.
    - **Results:** A brief analysis of the system's performance, explicitly noting the error between the estimated position and ground truth.
- **Simulation Results Plot:** Evidence of the functioning simulation. This must be either:
    - Submit a plot in PNG or PDF format comparing your estimated trajectory with the simulator's ground-truth trajectory. Clearly label the axes and units and use a legend to distinguish the two trajectories.
    - You may also include a short demonstration video, but it is optional and is not required for assessment.
- **Source Code & Git History:**
    - The scripts or project files are used to run the simulation and perform the localization calculations.
    - The hidden *.git* folder must be included in the directory to prove iterative development. Code must contain at least three meaningful commits with descriptive messages.

4. Assessment Criteria

Your submission will be assessed using the criteria below. The weights add up to 100% of the assignment mark.

Assessment Category	Weight	Marking Criteria for Top Band (70%+)
Report: Problem & Methods	30%	Your report clearly defines the localization problem and explains your chosen method accurately. It identifies the data used to estimate the robot's pose and describes the assumptions about odometry and sensor uncertainty.
Report: Results Analysis	20%	Your report compares the estimated pose with ground truth and discusses the accuracy of your results. It explains sources of error and any effects of drift or noise, using evidence from your simulation.
Simulation Results Plot	30%	Your plot clearly shows the estimated and ground-truth trajectories. The axes and units are labelled, and a legend distinguishes the two trajectories. The plot provides clear evidence of the localization performance.
Source Code and Git History	20%	Your code is organized and readable. Your submission includes the *.git* folder, with at least three meaningful commits that record your development process and use descriptive messages to explain the changes made.